

Foundation over Frontier: The Age of Science Cited by Patents after the Diffusion of Generative AI

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Abstract

The scientific base of corporate invention shifted after generative AI spread through corporate R&D, but not in the direction the acceleration argument predicts. We ask whether the post-2022 diffusion of generative AI (GenAI) changed the age of the science that patents build on, and whether it shortened the lag between academic publication and corporate patent citation. Using 441,537 patent–paper citation links from the Lens covering US filings from 2018 to 2025, we estimate a difference-in-differences design: technology classes with high GenAI exposure (CPC G06N, G16B, G16C, C40B) form the treated group, mature mechanical classes (F16B, F16H, B65D) form the control group, and the treatment period begins in 2023. We report three findings. First, the share of citations to papers three years old or younger fell from 20.6% to 11.9% in treated classes (a patent-level difference-in-differences of -5.6 percentage points), and the citation mass moved into the four-to-eight-year window (2015–2019 publications for 2023–2024 filings), not into the oldest stock. The claim is distributional and we hold to that: the mean, median and citation-weighted age of the cited literature do not move detectably, so exposed patenting became less concentrated on very recent science rather than drawing on older science. The estimate survives an event study with a 2023 break, science-citing alternative control classes, an independent citation source (Reliance on Science), a relaxed dating rule that recovers 67% more links, and within-firm estimation. Second, decomposing the decline shows where it lives: roughly half is carried by patents that themselves carry generative-model CPC codes, and firms that adopt generative patenting retain recent science in their other patents *better* than non-adopters, so the shift primarily reflects a reorientation of inventive output toward generative models built on the deep-learning canon of 2015–2019. Third, the citation-weighted lag lengthened (about $+0.21$ log points), but this secondary result halves against science-citing controls and disappears within firms: a composition effect across applicants, not a slowdown of the typical patent. A placebo estimation with a false 2017 treatment on 2015–2019 data shows no lag effect, while fresh-science intake in treated classes was rising before 2023: the decline is a reversal. In the first technology classes to absorb generative AI the scientific foundation of corporate invention moved away from the frontier and toward the canon, and the movement is carried by the turn to generative-model patenting itself; the acceleration expectation that the same design tests against this one is not supported in any subsample.

Keywords

generative AI, science of science, science-to-technology lag, patent citations, difference-in-differences, knowledge flows

1. Introduction and Motivation

Corporate R&D depends on academic science: firms read the literature, absorb new findings, and translate them into technologies that surface as patents. This translation is slow. Prior work has documented a science-to-technology lag that typically spans years between the publication of a scientific result and its citation in a patent [1, 2]. In biomedicine, patent citations trail scientific citations by a median of six years [7], and the time to first patent citation averages roughly a decade, varying with basicness and novelty [8].

After 2023, generative AI (GenAI) gave R&D teams tools to synthesize scientific knowledge, automate code, and screen literature at scale. Whether this improves corporate innovation is open: GenAI may help firms connect distant fields, or rivals relying on the same foundation models may converge on faster but less distinctive, less disruptive patents [3, 4].



2. Research Questions

- **RQ1 (Speed).** Has GenAI adoption after 2023 shortened the lag between academic publication and corporate patent citation?
- **RQ2 (Homogenization, descriptive).** Do patent abstracts in GenAI-exposed classes show semantic convergence across firms after 2023?

3. Data and Methods

Data. We draw patent records and their citations to scholarly works from the Lens,¹ which links patents to the papers they cite with dated records on both sides, complemented by OpenAlex for paper-level attributes [5]. The sample covers filings from 2018 through 2025 in seven CPC classes selected from the official USPTO/EPO definitions: a GenAI-exposed treated group (G06N, artificial intelligence and machine learning; G16B, bioinformatics; G16C, computational chemistry and materials; C40B, combinatorial chemistry) and a low-exposure mechanical control group (F16B, fastening devices; F16H, gearing; B65D, containers and packaging).

Identification. Difference-in-differences on the citation links: the log lag (days from paper publication to patent filing) and the patent-level share of citations to papers ≤ 3 years old are regressed on treated class $\times$ post-2023 filing, with class and filing-year fixed effects and standard errors clustered by class-year. Field-wide shocks common to both groups, such as examination speed-ups or drifting citation practices, are absorbed.

Results. The patent-level share of fresh science (≤ 3 years old) falls by 5.6 percentage points in treated classes ($p = 0.0004$; from 20.6% to 11.9%), robust to restricting filings to mid-2024. Shifted windows locate where the mass went: the four-to-eight-year share rises (+8.5 points, $p = 0.016$) while the share of papers eight years or older does not (−5.2 points, n.s.): the shift is toward 2015–2019 publications, not the oldest stock. An event study shows a break in 2023 (−3.7) that deepens in 2024 (−7.1, $p = 0.028$). Dropping the threshold altogether disciplines the claim: patent-level mean age (+0.14 y, $p = 0.86$), median age (+0.35, $p = 0.64$) and citation-weighted age (+0.39, $p = 0.60$) are all null, so what moved is the shape of the age distribution and not its centre. The result replicates against science-intensive alternative control classes (G01N, G02B, B01D, G21: −4.8 points, on top of a broad decline those classes themselves experience), on an independent citation source (Reliance on Science, −11.9 points on common patents), and strengthens to −10.1 when the dating rule is relaxed to publication year, recovering 67% more links. Decomposition attributes the decline: roughly half is carried by patents with generative-model CPC codes (excluding them leaves −2.7, $p = 0.09$), and adopting firms retain fresh science in their non-generative patents better than non-adopters (+4.5 points, $p = 0.021$), so the effect is primarily a shift in the direction of invention. The treated $\times$ post coefficient on the pair-level log lag is +0.21 ($p < 0.001$) under a uniform ten-year window, but it halves against science-citing controls and disappears within firms: a between-applicant composition effect, with the median patent’s lag unchanged.

Robustness and inference. A placebo estimation on 2015–2019 with a false 2017 treatment date yields no lag effect ($p = 0.44$). On the composition margin the fresh-science share was rising in treated classes before 2023 (+13.6 points in the placebo window), which rules out field maturation but means pre-trends were not parallel; the event study above is therefore the primary timing test. Because treatment is assigned at the level of seven CPC classes, class-level inference is limited by construction: the magnitude is stable across specifications while the uncertainty is substantial. Wild cluster bootstrap p-values for the share are 0.09–0.15 depending on the cluster set, and randomization inference over all 35, 126 and 715 possible class assignments agrees (0.143, 0.103, 0.090); a relative-magnitudes sensitivity analysis [9] gives breakdown values near $M = 0.75$, so the event-study coefficients do not survive a violation of parallel trends the size of the worst

¹<https://www.lens.org>

one the pre-period itself shows. The reading rests on the convergence of quasi-independent tests that do not share the seven-class structure, above all the contrast inside G06N, rather than on any single p-value. For RQ2, VOSviewer term maps [10] of G06N abstracts before and after 2023 show a vocabulary shift toward generative-model terms but no collapse into fewer clusters, and an exploratory cross-firm lexical-similarity analysis shows no rise. We acknowledge the limitations of direct citation linkage [11]; we retain it because our question is temporal and direct citations are the only linkage yielding two dated events.

4. Contributions

For corporate strategy, the shift is concentrated in the patents that embody the technology rather than spread across the firms that adopt it, and adopters retain fresh science better elsewhere, so absorptive capacity for frontier literature remains a differentiator rather than something the tools supply. For the science of science, the results link the GenAI shock to the debate on declining disruptiveness [4]: a reallocation of patent citations away from the newest science, driven primarily by the reorientation of invention toward generative models. Archive:²

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